# Adaptive Tutoring System

## Complete Platform Architecture

## & Functionality Reference

Version 0.1.0

Monorepo: NestJS API + React Frontend + FastAPI Worker

PostgreSQL | Redis | MinIO | Docker

# 1. Project Overview

Name: adaptive-tutoring-system (v0.1.0)

Type: Polyglot monorepo with 3 services + 2 shared packages

Purpose: AI-powered adaptive tutoring platform with knowledge component modeling, LLM-integrated content/question generation, assessment delivery, and mastery tracking using Bayesian Knowledge Tracing (BKT).

# 2. Tech Stack

| Layer | Technologies |
|---|---|
| Frontend | React 18.3, Vite 5.4, TypeScript 5.5, Tailwind CSS 4.1, Reac |
| Backend API | NestJS 10.4, TypeScript 5.5, Prisma 6.19, PostgreSQL 16, Red |
| Worker | FastAPI 0.115, Python 3.12, Uvicorn, psycopg2, Pydantic 2.9 |
| Storage | MinIO (S3-compatible blob storage) |
| DevOps | Docker Compose, GitHub Actions CI/CD, TurboRepo 2.3, pnpm 10 |

# 3. Monorepo Structure

```
capstone/
+-- apps/
|   +-- api/          # NestJS backend (port 3000) - 24 modules, 132+ endpoints
|   +-- web/          # React frontend (port 5173) - 17 routes, 35+ components
|   +-- worker/       # FastAPI worker (port 8000) - async grading + mastery
+-- packages/
|   +-- shared/       # Zod schemas & TypeScript types (35+ schemas)
|   +-- config/       # ESLint + Prettier shared configs
+-- e2e/              # Playwright E2E tests
+-- infra/            # Infrastructure scripts
+-- docker-compose.yml  # Full stack definition
```

# 4. Database Schema (PostgreSQL + Prisma ORM)

## 4.1 Enums (14 total)

- **UserRole:** student, teacher, admin
- **QuestionType:** text, drawing, mixed, MCQ_SINGLE, MCQ_MULTI, STRUCTURED
- **AttemptStatus:** in_progress, submitted, grading, graded
- **GradedBy:** auto, manual
- **EventStatus:** pending, processing, processed, failed
- **CourseStatus:** DRAFT, PUBLISHED
- **CourseVisibility:** PUBLIC, PRIVATE
- **ModuleItemType:** PAGE, PDF, LINK, ASSESSMENT
- **EnrollmentStatus:** ACTIVE, DROPPED
- **DraftStatus:** DRAFT, APPROVED, REJECTED
- **ChunkingStrategy:** FIXED_SIZE, PARAGRAPH, SEMANTIC
- **GenerationJobStatus:** PENDING, RUNNING, COMPLETED, FAILED
- **EvaluationRunStatus:** PENDING, RUNNING, COMPLETED, FAILED
- **ProposedKCStatus:** PROPOSED, APPROVED, ARCHIVED

## 4.2 Core Models (30+ tables)

### Authentication & Users

- **User:** id, email, passwordHash, name, role, llmProvider, encryptedApiKey, llmModel, createdAt, updatedAt

### Course Structure

- **Course:** id, title, description, teacherId, status (DRAFT/PUBLISHED), visibility (PUBLIC/PRIVATE), bannerBlobKey, orderIndex, archivedAt (soft delete)
- **Topic:** id, courseId, title, description, orderIndex, generationJobId
- **CourseModule:** id, courseId, topicId, title, orderIndex, generationJobId
- **ModuleItem:** id, moduleId, type (PAGE/PDF/LINK/ASSESSMENT), title, orderIndex, contentMdx, pdfBlobKey, pdfFilename, pdfSize, url, assessmentId, learningOutcomes (JSON), estimatedMinutes
- **PageContentVersion:** id, itemId, version, contentJson (BlockDocument JSON), source (human/ai/rollback), authorId, jobId

### Question Bank & Assessments

- **Question:** id, courseId, topicId, version, prompt, stem (JSON rich blocks), type, maxScore, options (JSON MCQ), correctOptionId, correctAnswer (JSON), explanation (JSON), difficulty (1-5), bloomsLevel, kcIds (UUID[]), pageIds (UUID[]), tags, status, createdBy, sourceType, provenance
- **Assessment:** id, courseId, title, description, mode (practice/test), isPublished, settings (JSON: timeLimit, maxAttempts, showSolutions, shuffle)
- **AssessmentQuestion:** id, assessmentId, questionId, orderIndex, points, section

### Student Attempts & Grading

- **Attempt:** id, studentId, questionId, assessmentId, status, textResponse, selectedOptionIds (JSON), strokesJson (JSON), drawingBlobUrl, workingStrokes (JSON), workingBlobUrl, currentScore, autoFeedback, submittedAt
- **GradingResult:** id, attemptId, score, feedback, gradedBy (auto/manual), graderId

### Enrollment

- **Enrollment:** id, studentId, courseId, status (ACTIVE/DROPPED), enrolledAt. Unique: studentId+courseId

### Knowledge Components (KC)

- **KnowledgeComponent:** id, topicId, code, label, description (legacy model)
- **ProposedKC:** id, courseId, name, description, status (PROPOSED/APPROVED/ARCHIVED), confidenceLevel (LOW/MEDIUM/HIGH), createdBy (ai/human), generationJobId, relatedToKCId, evidence (JSON), pageIds, topicId, subtopicId
- **KcEdge:** id, courseId, fromKcId, toKcId, relationship (prerequisite/related/builds_on), weight, createdBy
- **KcContentMapping:** id, courseId, kcId, pageId, strength (LOW/MEDIUM/HIGH), blockIds, createdBy
- **QuestionKc:** id, questionId, kcId, weight
- **KcEvidence:** id, gradingResultId, kcId, isCorrect, score

### Mastery Tracking (BKT-Ready)

- **UserMastery:** id, userId, kcId, probabilityKnown (default 0.5), totalAttempts, correctAttempts, lastAttemptAt, pLearn, pGuess, pSlip, pTransit

### RAG & Document Management

- **SourceDocument:** id, courseId, uploadedById, title, filename, blobKey, mimeType, sizeBytes, pageCount, chunkingStrategy, chunkCount, indexedAt
- **DocumentChunk:** id, documentId, chunkIndex, content, pageNumber, tokenCount, embedding (JSON)

### LLM Content Generation

- **ContentDraft:** id, courseId, createdById, title, contentMdx, citations (JSON), status (DRAFT/APPROVED/REJECTED), version, parentDraftId
- **LlmGenerationJob:** id, courseId, userId, status, jobType, inputPayload, outputPayload, retrievedChunkIds, errorMessage, model, promptTokens, completionTokens, durationMs
- **LlmAuditLog:** id, draftId, courseId, userId, action, model, promptTokens, completionTokens, durationMs, inputPayload, outputPayload

### Evaluation & Quality Assurance

- **EvaluationRun:** id, courseId, userId, status, config (JSON), summary (JSON)
- **PageEvalResult:** id, runId, itemId, itemTitle, scores (JSON: formatting/equations/pedagogy/rigor/overall), issues (JSON array), mathIssues, fixesApplied
- **KcEvaluationRun:** id, courseId, userId, status, scope (JSON), depth (quick/deep), results (JSON)
- **CurriculumCoverageRun:** id, courseId, userId, status, syllabusText, parsedOutcomes, outcomeKcMaps, coverageResults
- **PublishGateRun:** id, courseId, userId, checks (JSON), overallPass, overridden, overrideReason, published
- **KnowledgeVersion:** id, courseId, version, changeSummary, changeType, source, authorId, snapshotKcs, snapshotEdges, snapshotMaps

### Visualization & Events

- **KcGraphLayout:** id, courseId (unique), layoutJson (JSON: nodes with x/y, viewBox)
- **EventQueue:** id, topic, payload (JSON), status, processedAt

# 5. Complete API Endpoints (132+)

## 5.1 Auth (3 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /auth/register | Public | Register user (throttled 5/min) |
| POST | /auth/login | Public | Login (throttled 10/min) |
| GET | /auth/me | JWT | Get current user |

## 5.2 Courses (9 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /courses | teacher | Create course |
| GET | /courses | Any | List courses (role-filtered) |
| GET | /courses/catalog | Any | Public course catalog |
| GET | /courses/:id | Any | Get course details |
| PATCH | /courses/:id | teacher | Update course |
| POST | /courses/:id/duplicate | teacher | Deep clone course |
| POST | /courses/:id/publish | teacher | Publish course |
| POST | /courses/:id/unpublish | teacher | Unpublish course |
| DELETE | /courses/:id | teacher | Soft delete course |

## 5.3 Topics (5 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /topics | teacher | Create topic |
| GET | /topics | Any | List topics (query: courseId) |
| GET | /topics/:id | Any | Get topic |
| PATCH | /topics/:id | teacher | Update topic |
| DELETE | /topics/:id | teacher | Delete topic |

## 5.4 Modules (4 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /courses/:cId/modules | teacher | Create module |
| PATCH | /courses/:cId/modules/:id | teacher | Update module |
| DELETE | /courses/:cId/modules/:id | teacher | Delete module |
| POST | /courses/:cId/modules/reorder | teacher | Reorder modules |

### 5.5 Module Items (9 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /modules/:mId/items | teacher | Create item (PAGE/PDF/LINK/ASSESSMENT) |
| PATCH | /modules/:mId/items/:id | teacher | Update item |
| DELETE | /modules/:mId/items/:id | teacher | Delete item |
| POST | /modules/:mId/items/reorder | teacher | Reorder items |
| POST | /items/:id/upload-url | teacher | Get S3 presigned upload URL |
| GET | /items/:id/download-url | Any | Get download URL |
| GET | /items/:id/versions | teacher | Get version history |
| GET | /items/:id/versions/:vId | teacher | Get specific version |
| POST | /items/:id/versions/:vId/rollback | teacher | Rollback to version |

### 5.6 Enrollments (6 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /enrollments | teacher | Enroll student |
| POST | /enrollments/self | student | Self-enroll in public course |
| POST | /enrollments/:cId/drop | student | Drop course |
| GET | /enrollments | teacher | List enrollments |
| GET | /enrollments/my | student | My enrollments |
| DELETE | /enrollments/:id | teacher | Remove enrollment |

### 5.7 Questions (10 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /questions | teacher | Create question |
| GET | /questions | teacher | List (filters: course,topic,type,status,etc) |
| GET | /questions/course/:cId | Any | Course questions |
| POST | /questions/bulk-import | teacher | Bulk import |
| POST | /questions/bulk-status | teacher | Bulk status update |
| GET | /questions/:id | Any | Get question |
| PATCH | /questions/:id | teacher | Update question |
| DELETE | /questions/:id | teacher | Delete question |
| POST | /questions/:id/duplicate | teacher | Duplicate question |
| POST | /questions/:id/archive | teacher | Archive question |

### 5.8 AI Question Generation (2 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /questions/generate | teacher | Generate via LLM+RAG |
| POST | /questions/generate/save | teacher | Save generated questions to bank |

### 5.9 Assessments (10 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /assessments | teacher | Create assessment |
| GET | /assessments | Any | List assessments |
| GET | /assessments/available | student | Available assessments |
| GET | /assessments/:id | Any | Get assessment |
| PATCH | /assessments/:id | teacher | Update assessment |
| DELETE | /assessments/:id | teacher | Delete assessment |
| POST | /assessments/:id/questions | teacher | Add question |
| DELETE | /assessments/:id/questions/:qId | teacher | Remove question |
| POST | /assessments/:id/reorder | teacher | Reorder questions |
| PATCH | /assessments/:id/questions/:qId | teacher | Update question item |

### 5.10 Attempts (8 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /attempts | student | Create attempt |
| GET | /attempts | teacher | List all attempts |
| GET | /attempts/my | student | My attempts |
| GET | /attempts/review | teacher | Pending review |
| GET | /attempts/:id | Any | Get attempt |
| PATCH | /attempts/:id | student | Update attempt (auto-save) |
| POST | /attempts/:id/submit | student | Submit attempt |
| POST | /attempts/:id/grade | teacher | Manual grade |

### 5.11 Blob Storage (5 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| GET | /blobs/presign/upload | Public | Presigned S3 upload URL |
| GET | /blobs/presign/download | Public | Presigned S3 download URL |
| PUT | /blobs/:aId/:type | Public | Upload blob |
| GET | /blobs/:aId/:type | Public | Download blob |
| DELETE | /blobs/:aId/:type | Public | Delete blob |

### 5.12 RAG & LLM (13 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| GET | /courses/:cId/documents | teacher | List documents |
| POST | /courses/:cId/documents | teacher | Upload document |
| DELETE | /documents/:dId | teacher | Delete document |
| POST | /documents/:dId/rechunk | teacher | Rechunk document |
| POST | /courses/:cId/rag/query | teacher | RAG query |
| POST | /courses/:cId/rag/generate | teacher | Generate content draft |
| GET | /courses/:cId/drafts | teacher | List drafts |
| GET | /drafts/:dId | teacher | Get draft |
| POST | /drafts/:dId/approve | teacher | Approve draft |
| POST | /drafts/:dId/reject | teacher | Reject draft |
| GET | /courses/:cId/llm-logs | teacher | LLM audit logs |
| GET | /llm-settings | teacher | Get LLM settings |
| POST | /llm-settings | teacher | Save LLM API key/model |

### 5.13 AI Course Structure (4 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /admin/courses/:cId/generate-structure | teacher | Generate structure (curriculum/level based) |
| POST | /admin/courses/:cId/apply-structure/:jId | teacher | Apply generated structure |
| GET | /admin/courses/:cId/structure-jobs | teacher | List jobs |
| GET | /admin/courses/:cId/structure-jobs/:jId | teacher | Get job |

### 5.14 AI Page Content (3 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /admin/ai/suggest-prompt | teacher | Get prompt suggestion |
| POST | /admin/pages/:pId/generate-content | teacher | Generate single page |
| POST | /admin/pages/generate-content-batch | teacher | Batch generate (max 20) |

### 5.15 Content Evaluation (5 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| GET | /courses/:cId/evaluation/tree | teacher | Page tree for selector |
| POST | /courses/:cId/evaluation/run | teacher | Start evaluation |
| GET | /courses/:cId/evaluation/runs | teacher | List runs |
| GET | /courses/:cId/evaluation/runs/:rId | teacher | Get run results |
| POST | /courses/:cId/evaluation/apply-fixes | teacher | Apply auto-fixes |

### 5.16 Mastery & KC Legacy (5 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| GET | /me/mastery | student | My mastery by topic |
| GET | /students/:id/mastery | teacher | Student mastery |
| POST | /me/revise-worksheet | student | Generate revision worksheet |
| POST | /kcs | teacher | Create KC |
| GET | /kcs | teacher | List KCs |

### 5.17 Proposed KCs (10 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| GET | /proposed-kcs | teacher | List KCs (courseId, status) |
| GET | /proposed-kcs/stats | teacher | KC stats |
| GET | /proposed-kcs/similar-pairs | teacher | Find duplicates |
| GET | /proposed-kcs/health | teacher | Health indicators |
| GET | /proposed-kcs/:id | teacher | Get KC |
| PATCH | /proposed-kcs/:id | teacher | Update KC |
| POST | /proposed-kcs/:id/approve | teacher | Approve KC |
| POST | /proposed-kcs/:id/archive | teacher | Archive KC |
| POST | /proposed-kcs/merge | teacher | Merge KCs |
| DELETE | /proposed-kcs/:id | teacher | Delete KC |

### 5.18 KC Mappings (7 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| GET | /kc-mappings | teacher | List mappings |
| GET | /kc-mappings/by-kc/:kcId | teacher | By KC |
| GET | /kc-mappings/by-page/:pId | teacher | By page |
| GET | /kc-mappings/summary | teacher | Bidirectional summary |
| POST | /kc-mappings | teacher | Upsert mapping |
| PATCH | /kc-mappings/:id | teacher | Update strength |
| POST | /kc-mappings/auto-generate | teacher | Auto-generate from evidence |

### 5.19 KC Graph (12 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /kc-graph/generate | teacher | AI-generate graph |
| GET | /kc-graph | teacher | Full graph |
| POST | /kc-graph/edges | teacher | Add edge |
| PATCH | /kc-graph/edges/:id | teacher | Update edge |
| DELETE | /kc-graph/edges/:id | teacher | Remove edge |
| GET | /kc-graph/topological-order | teacher | Topological sort |
| GET | /kc-graph/cycles | teacher | Detect cycles |
| POST | /kc-graph/layout | teacher | Save layout |
| GET | /kc-graph/layout | teacher | Load layout |
| PATCH | /kc-graph/hierarchy/:kcId | teacher | Update KC hierarchy |
| POST | /kc-graph/hierarchy/batch | teacher | Batch hierarchy |
| GET | /kc-graph/hierarchy-options | teacher | Available topics/modules |

### 5.20 KC Evaluation (4 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /courses/:cId/kc-evaluation/run | teacher | Start KC evaluation |
| GET | /courses/:cId/kc-evaluation/runs/:rId | teacher | Get run |
| GET | /courses/:cId/kc-evaluation/runs | teacher | List runs |
| GET | /courses/:cId/kc-evaluation/runs/:rId/export-json | teacher | Export |

### 5.21 Curriculum Coverage (7 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /courses/:cId/curriculum-coverage/run | teacher | Start analysis (syllabusText) |
| GET | /courses/:cId/curriculum-coverage/runs | teacher | List runs |
| GET | /courses/:cId/curriculum-coverage/runs/:rId | teacher | Get run |
| POST | /courses/:cId/curriculum-coverage/runs/:rId/mapping | teacher | Update mapping |
| DELETE | /courses/:cId/curriculum-coverage/runs/:rId/mapping | teacher | Remove mapping |
| POST | /courses/:cId/curriculum-coverage/runs/:rId/reanalyz | teacher | Re-analyze |
| GET | /courses/:cId/curriculum-coverage/runs/:rId/export-js | teacher | Export |

### 5.22 Knowledge Versioning (6 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| GET | /courses/:cId/knowledge-versions | teacher | List versions |
| GET | /courses/:cId/knowledge-versions/:vId | teacher | Get snapshot |
| GET | /courses/:cId/knowledge-versions/:vId/diff | teacher | Diff vs current |
| GET | /courses/:cId/knowledge-versions/compare | teacher | Diff two versions |
| POST | /courses/:cId/knowledge-versions/:vId/restore | teacher | Restore version |
| GET | /courses/:cId/knowledge-versions/:vId/export | teacher | Export |

### 5.23 Publish Gate (4 endpoints)

| Method | Path | Roles | Description |
|---|---|---|---|
| POST | /courses/:cId/publish-gate/check | teacher | Dry run checks |
| POST | /courses/:cId/publish-gate/publish | teacher | Publish if pass |
| POST | /courses/:cId/publish-gate/publish-override | teacher | Force publish with reason |
| GET | /courses/:cId/publish-gate/audit | teacher | Audit history |

### 5.24 WebSocket Gateway

- **join:** Student joins room student:{studentId}
- **grade_completed:** Real-time grading notification to student with score/feedback

# 6. Frontend Architecture (React + Vite + TypeScript)

## 6.1 Routes (17 routes)

### Public Routes

- /login - Email/password login
- /register - Register with role selection (student/teacher)
- /health - Health check status page

### Teacher Routes (teacher, admin)

- /teacher - Dashboard with stat cards
- /teacher/courses - Course listing/management
- /teacher/courses/:courseId - Course builder (tabs: overview, content, sources, evaluate, knowledge, publish, settings)
- /teacher/studio/:courseId - Content studio with block editor
- /teacher/questions - Question bank (filter/create/AI generate)
- /teacher/assessments - Assessment builder
- /teacher/review - Student submission review
- /teacher/attempt/:attemptId - Grade individual attempt
- /teacher/ai-settings - LLM provider/key/model config

### Student Routes

- /student - Dashboard (mastery %, weak areas, revision worksheet)
- /student/catalog - Browse/enroll in courses
- /student/courses - My enrolled courses
- /student/courses/:courseId - View course content
- /student/assessments - Available assessments
- /student/attempt/:attemptId - Take assessment (text/drawing/MCQ)
- /student/results - View grades and feedback

## 6.2 State Management

- **AuthContext:** user state, token, login/register/logout, auto-validate via /auth/me, WebSocket room join for students
- **ToastContext:** success/error/info notifications, 4s auto-dismiss
- **Local state:** Heavy useState/useCallback/useMemo. No Redux/Zustand/Recoil.

## 6.3 Key Components (35+)

### Block Editor

TipTap-based rich content editor with drag-drop block reordering via dnd-kit. Block types: Text, Image, Video, Diagram, Callout, Divider. Auto-save capability.

### Drawing Canvas

HTML5 Canvas for drawing questions: pen/eraser tools, adjustable color/width, stroke history, undo/redo, PNG export.

### AI Generation Components

- **QuestionGenerationModal:** Config (type/quantity/difficulty/Blooms/KC/source) -> generating -> review
- **GenerateStructureWizard:** Source selection -> config -> preview -> apply
- **PromptComposerModal:** AI page content with scope preference (module_only/module_then_course/course_only)

### Knowledge Component Panels

- **KcStudioPanel:** KC CRUD, approve/archive, merge similar, health metrics, similarity detection
- **KcGraphStudioPanel:** Force-directed graph visualization, colored by subtopic, edge management
- **KcMappingPanel:** KC-Page mappings, auto-generate from evidence, add/edit strength
- **KcEvaluationDashboard:** KC evaluation metrics and coverage analysis
- **CurriculumCoveragePanel:** Syllabus-KC mapping, gap analysis, orphan/overrepresented KC detection
- **KnowledgeVersionPanel:** Version history, snapshots, diffs, restore
- **LearningPathPanel:** Visual learning path with prerequisites
- **PublishGatePanel:** Pre-publish checklist

### Other Components

- **MDXRenderer:** MDX compilation with KaTeX math + Mermaid diagrams
- **SourceSelector:** RAG source document selection
- **VersionHistoryPanel:** Page content version comparison
- **BulkProgressPanel:** Progress indicator for bulk operations

## 6.4 API Client & WebSocket

apiFetch<T>() - Core fetch wrapper with JWT injection, 401 auto-redirect, JSON parsing. Convenience: api.get/post/patch/delete/upload.

Socket.IO client connects on student login, joins student:{id} room, listens for grade_completed events for real-time score updates.

# 7. Worker Service (Python FastAPI)

## 7.1 Event Processing Flow

```
Poll PostgreSQL event_queue (every 2s) for 'grade_submission' events:
  1. Update attempt status -> 'grading'
  2. Grade attempt (string matching against rubric answer_key)
  3. Insert grading result into database
  4. Update attempt status -> 'graded'
  5. Update KC mastery using EMA (alpha=0.3)
  6. Publish 'grade_completed' event
  7. Mark original event as processed
```

## 7.2 Grading Logic

- MCQ: Auto-graded on NestJS backend before reaching worker
- Text/Structured: Case-insensitive string matching against rubric answer_key
- Drawing: Requires manual teacher review
- Missing rubric: Returns 'Manual review required'

## 7.3 Mastery Update (EMA Algorithm)

```
P(known) = alpha * current_score + (1 - alpha) * old_probability
alpha = 0.3, initial P = 0.5, correct threshold = 0.5

For each KC linked to the question:
  - Insert KC evidence record (append-only audit trail)
  - Update/Create UserMastery record with EMA formula
  - Increment totalAttempts and correctAttempts counters
```

# 8. Shared Package (@ats/shared)

35+ Zod schemas providing runtime validation and TypeScript types shared across the monorepo:

- **User schemas:** User, CreateUser, Login, AuthResponse
- **Course schemas:** Course, Topic, Question, MCQOption, Enrollment + Create/Update variants
- **Assessment schemas:** Assessment, AssessmentQuestion, Attempt, GradingResult, Stroke/StrokePoint + Create/Update/Submit variants, ManualGrade
- **Mastery schemas:** KnowledgeComponent, CreateKc, QuestionKc, UserMastery, KcEvidence, KcMasteryItem, MasteryByTopic
- **Question Generation:** GenerateQuestionsRequest/Response, GeneratedQuestion/Option/Tags/Provenance, SaveGeneratedQuestionsRequest
- **Event Bus:** GradeSubmissionPayload, GradeCompletedPayload
- **Enums:** UserRole, QuestionType, BloomsLevel, AttemptStatus, EventStatus, SourceMode, Difficulty

# 9. Security & Infrastructure

- JWT Auth with Passport.js, Bearer token from Authorization header
- Role-based access control via @Roles() decorator + RolesGuard
- Rate limiting via Throttler with Redis store (default 30/60s, auth: 5-10/min)
- ZodValidationPipe for request body validation
- GlobalExceptionFilter: HttpException, ZodError, PrismaClientKnownRequestError
- Cascade deletes on all foreign keys
- Soft deletes for courses (archivedAt), questions (status), KCs (status)
- Per-user encrypted LLM API keys stored in database
- Docker Compose for full stack: api, web, worker, postgres, redis, minio
- GitHub Actions CI/CD: lint, typecheck, integration tests, E2E tests
- TurboRepo for build orchestration with dependency-aware caching

# 10. Key Architectural Patterns

- **Event-driven async processing:** PostgreSQL event queue with polling worker for decoupled grading
- **Knowledge graph modeling:** ProposedKCs with edges (prerequisite/related/builds_on), topological sort, cycle detection
- **Version control:** Page content versions, knowledge graph snapshots with diff/restore
- **RAG pipeline:** Document upload -> chunking (fixed/paragraph/semantic) -> embedding -> retrieval -> LLM generation
- **BKT-ready mastery:** EMA-based probability updates with BKT parameters (pLearn, pGuess, pSlip, pTransit)
- **Block-based content:** Structured JSON blocks (text, image, video, diagram, callout, divider)
- **Multi-step AI workflows:** Course structure -> page content -> evaluation -> publish gate
- **Audit trail:** LLM audit logs tracking all AI calls with token usage and duration
- **Monorepo code sharing:** Shared Zod schemas + TypeScript types consumed by all apps via workspace packages